LOGOS V1

Logical Optimization for Global Outcome Systems

A system for detecting logical inconsistencies in prediction markets

LOGOS (Logical Optimization for Global Outcome Systems) *is a research system that analyzes prediction markets to test whether their implied probabilities obey basic logical constraints. Prediction market prices are often interpreted as probabilities, but related contracts, such as price thresholds or nested events, must satisfy certain mathematical relationships. LOGOS organizes markets into structured families, detects violations of these probability constraints, and computes the nearest logically coherent probability system using convex optimization. The system then measures how frequently these inconsistencies occur, how large they are, and how long they persist. Rather than predicting outcomes, LOGOS studies the* ***internal coherence of market beliefs****, offering insight into structural inefficiencies in prediction markets.*

1. Project Overview

We are building **LOGOS**, a research prototype that analyzes prediction markets and tests whether their implied probabilities obey basic logical and probabilistic constraints.

Prediction markets are often interpreted as aggregating information into probabilities. However, markets frequently contain sets of related contracts whose prices should obey simple logical relationships. In practice, those relationships are often violated.

LOGOS detects these violations and computes the **nearest logically coherent probability system** using convex optimization.

Rather than predicting market outcomes, the goal is to study **whether the market's internal probability structure is coherent**.

This project sits at the intersection of:

- probability theory
- convex optimization
- market microstructure
- empirical financial analysis

The objective is to produce a **clean research prototype** rather than a generic trading bot or prediction system.

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2. Motivation

Most undergraduate quant projects focus on:

- predicting stock prices
- basic ML models on returns
- simple factor backtests
- toy option pricing implementations

These are common and often shallow.

Instead, LOGOS focuses on **market structure and logical consistency**.

Prediction markets claim to represent probabilities. If that is true, then those probabilities must satisfy certain mathematical constraints. For example:

If the market says:

- P(BTC > 95k by June) = 0.48
- P(BTC > 100k by June) = 0.52

this is logically impossible, since exceeding 100k implies exceeding 95k.

LOGOS is designed to detect these kinds of violations systematically and quantify how far market prices deviate from logical coherence.

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3. Data Source

Version 1 will use data from **Polymarket**, a prediction market platform where contracts trade between 0 and 1 and represent probabilities of future events.

Polymarket is well suited for this project because:

- prices map directly to probabilities
- markets often have naturally related contracts
- event-based contracts create clear logical relationships

Examples include:

- price thresholds (BTC above X)
- event deadlines (event by March vs by June)
- hierarchical achievements (team reaches finals vs wins championship)
- mutually exclusive outcomes (candidate A vs candidate B)

These natural relationships allow LOGOS to organize markets into structured groups.

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4. Core Idea

LOGOS operates in four conceptual steps:

1. **Collect market probabilities**
2. **Organize related markets into logical families**
3. **Check whether the probabilities violate logical constraints**
4. **Compute the nearest coherent probability system**

The key mathematical step is projecting observed probabilities onto the **nearest logically consistent probability vector**.

This projection is computed using convex optimization.

The result allows us to quantify:

- how often logical violations occur
- how large they are
- how long they persist

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5. Project Scope

The project will be built in **two conceptual stages**, but only the first will be implemented in the initial timeline.

Version 1 (4 weeks)

LOGOS will include:

- Polymarket data ingestion
- market parsing and structuring
- grouping contracts into logical families
- constraint detection
- convex optimization to compute coherent probabilities
- empirical analysis of violations

Version 2 (future work)

Possible extensions include:

- automatic dependency detection using language models
- real-time violation monitoring
- trade simulation / backtesting
- cross-platform arbitrage analysis

- Bayesian probability updating

For now, the goal is to build a **clean and reliable core system**.

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6. Market Family Types

LOGOS organizes markets into **families**, where contracts share logical relationships.

Version 1 will support four family types.

6.1 Threshold Chains

Example:

- BTC above 95k by June
- BTC above 100k by June
- BTC above 105k by June

Constraint:

$P(BTC > 95k) \geq P(BTC > 100k) \geq P(BTC > 105k)$

Higher thresholds must have lower probability.

This is the **primary market family** because it is clean and common.

6.2 Deadline Nesting

Example:

- BTC above 100k by March

- BTC above 100k by June
- BTC above 100k by December

Constraint:

P(by March) ≤ P(by June) ≤ P(by December)

Later deadlines must be at least as likely.

This introduces **time structure** into the logic system.

6.3 Nested Achievement Chains

Example:

- team makes playoffs
- team reaches conference finals
- team wins championship

Constraint:

P(championship) ≤ P(finals) ≤ P(playoffs)

More difficult achievements cannot be more likely than prerequisite ones.

This allows LOGOS to analyze more general event hierarchies.

6.4 Mutually Exclusive Outcome Sets

Example:

- Candidate A wins
- Candidate B wins

- Candidate C wins

Constraint:

$P(A) + P(B) + P(C) = 1$
(if the set is exhaustive)

or

$P(A) + P(B) + P(C) \leq 1$
(if not exhaustive)

This family tests **probability completeness and exclusivity**.

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7. Mathematical Framework

For each market family, we observe a vector of market probabilities:

p_{market}

LOGOS computes the nearest coherent probability vector:

p^*

This is solved as an optimization problem:

minimize:

$\Sigma (p_i - p_{market\_i})^2$

subject to:

- family-specific logical constraints
- $0 \leq p_i \leq 1$

This projection finds the **closest logically consistent belief system** to the observed market prices.

The optimization will be implemented using **CVXPY**.

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8. Key Metrics

The project focuses on three empirical questions.

Frequency

How often do market families violate logical constraints?

Example metrics:

- fraction of families with violations
- violations by family type
- violations over time

Magnitude

How large are the inconsistencies?

Example metrics:

- average adjustment size
- maximum deviation
- family-level inconsistency score

One measure:

$||p\_market - p^*||_2$

This captures how far the observed market system is from coherence.

Persistence

How long do violations last?

Example metrics:

- average violation duration
- median persistence
- persistence by market type

Persistence matters because short-lived violations may simply reflect noise, while long-lived ones suggest structural inefficiency.

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9. System Architecture

LOGOS will consist of several modules:

1. **Data ingestion**
   Fetch Polymarket market data
2. **Market parser**
   Extract structured fields (asset, threshold, deadline)
3. **Family builder**
   Group markets into logical families
4. **Constraint engine**
   Construct linear constraints for each family
5. **Optimization engine**
   Compute coherent probability projections
6. **Metrics engine**
   Measure violation frequency, magnitude, and persistence
7. **Visualization / analysis layer**
   Produce figures and summary statistics

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10. Division of Work

Caleb

- project framing and research design
- Polymarket data ingestion
- market family construction
- visualization
- empirical interpretation
- documentation and write-up

Northwestern collaborator

- constraint formulation
- optimization implementation
- CVXPY projection engine
- numerical analysis

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- debugging
- empirical analysis
- result validation
- system design decisions

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11. Success Criteria

LOGOS V1 is successful if it demonstrates that:

1. prediction markets sometimes violate logical probability constraints
2. these violations can be systematically detected
3. coherent probability systems can be computed via optimization
4. violations can be measured empirically in terms of frequency, magnitude, and persistence

Even if violations are rare, the project still provides a rigorous framework for analyzing market coherence.

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12. Why This Project Is Interesting

LOGOS shifts the focus away from “predicting markets” toward something deeper:

testing whether markets obey their own probabilistic logic.

This reframing highlights:

- structural inefficiencies
- probability incoherence
- logical arbitrage opportunities

It also demonstrates a combination of mathematical reasoning, systems design, and empirical analysis that goes beyond most standard undergraduate quant projects.